BLOCK 19

Dictionaries & Structured Records

Keys, values, and real-world data

“Look it up by name.”

```
record = {
  "record_id": "EQ-1045",
  "status": "active",
  "hours": 850
}
```

Instead of remembering positions,
we give values meaningful labels.

Today’s mental model

Dictionary = a labeled record
like a spreadsheet row, JSON object,
or API message.

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Why dictionaries matter

Lists are great — until the meaning depends on memory.

```
record = ["EQ-1045", "active", 850, True]

print(record[2])
```

What is position 2?
Hours? Temperature? Errors?

```
record = {
  "record_id": "EQ-1045",
  "status": "active",
  "operating_hours": 850,
  "inspection_complete": True,
}

print(record["operating_hours"])
```

A key explains what the value means.

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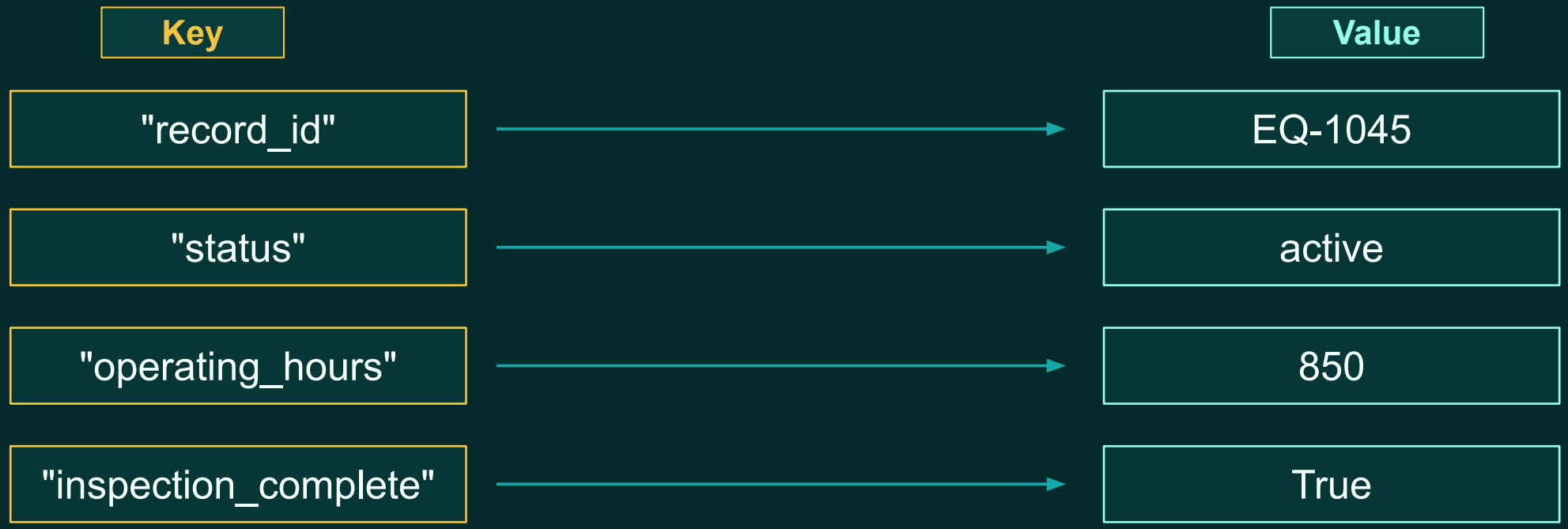
From positions to names

A dictionary connects a key to a value.



The key is not just a label on a slide.
It is how the program finds the value.

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Creating a dictionary

Curly braces hold named values.

```
equipment = {
  "equipment_id": "EQ-1045",
  "status": "active",
  "operating_hours": 850,
  "maximum_hours": 1000,
  "inspection_complete": True,
  "temperature": 72.5,
}
```

Syntax landmarks

- curly braces { }
- string keys
- colon between key and value
- comma between pairs
- mixed value types are allowed

Readable data beats mystery positions.

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Retrieving values

Use the key inside square brackets.

```
print(equipment["equipment_id"])
print(equipment["status"])
print(equipment["temperature"])
```

Output

```
EQ-1045
active
72.5
```

Mental model

Dictionary lookup is like asking:
“Give me the value stored under this label.”

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Updating and adding values

Dictionaries are mutable: the record can change.

```
equipment["status"] = "inactive"
```

Updates an existing key

```
equipment["error_count"] = 0
```

Adds a new key if it does not exist

Same square bracket syntax.
Different result depends on whether the key already exists.

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Removing keys and checking size

Records can also lose fields.

```
removed_value = equipment.pop("temperature")

print(removed_value)
print(len(equipment))
```

.pop() removes a key and returns its value.

len(dictionary) tells how many key-value pairs are present.

Common use

Removing temporary fields, optional fields, or data that should not be carried forward.

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Checking for keys

Before you look up a value, check that it exists.

```
has_status = "status" in equipment

if has_status:
    print(equipment["status"])
```

The in operator checks keys, not values.

```
"active" in equipment
```

False in most cases:
"active" is a value, not a key.

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Safe retrieval with .get()

Two ways to ask for a value.

```
equipment["error_count"]
```

Use when the key should exist.

```
equipment.get("error_count")
```

Returns None if the key is missing.

```
equipment.get("error_count", 0)
```

Returns 0 if the key is missing.

Default values are useful for optional fields.

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Dictionary methods

Three views of the same record.

```
record.keys()
record.values()
record.items()
```

keys()	values()	items()
field names	stored data	key-value pairs

You will use these heavily with loops.

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Dictionaries as structured records

This looks like real data.

```
employee = {
  "employee_id": "EMP-203",
  "name": "Jordan Smith",
  "department": "engineering",
  "active": True,
}
```

Same idea appears as:

- a row in a spreadsheet
- a JSON object
- a database record
- a message from an API

Dictionaries are a bridge from beginner Python
to real data-processing work.

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A spreadsheet row becomes a dictionary

Column names become keys.

employee_id	name	department	active
EMP-203	Jordan Smith	engineering	True

```
employee = {
  "employee_id": "EMP-203",
  "name": "Jordan Smith",
  "department": "engineering",
  "active": True,
}
```

A dictionary can represent one row.
A list of dictionaries can represent many rows.

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Nested data structures

A dictionary can contain another dictionary.

```
record = {
  "record_id": "EQ-1045",
  "status": "active",
  "metrics": {
    "temperature": 72.5,
    "operating_hours": 850,
  },
}
```

```
temperature = record["metrics"]["temperature"]
```

Read from left to right:

record → metrics → temperature

Nested data shows up in APIs and JSON files.

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Readiness checks inside a dictionary

Use named fields to make decisions.

```
equipment["is_ready"] = (
  equipment["status"] == "active"
  and equipment["inspection_complete"]
  and equipment["operating_hours"]
      <= equipment["maximum_hours"]
)
```

Result stored back into the same record.

The dictionary can hold both raw fields and derived fields.

This is the same validation logic from Block 16 — now attached to a structured record.

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Guided exercise 1: equipment record

Create, read, update, and add fields.

```
equipment = {
  "equipment_id": "EQ-1045",
  "status": "active",
  "operating_hours": 850,
  "maximum_hours": 1000,
  "inspection_complete": True,
  "temperature": 72.5,
}
```

Student tasks

- print the equipment ID
- print current status
- update operating hours
- add error_count
- add is_ready

Exercise time: 8 minutes

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Guided exercise 1: readiness field

Print every intermediate idea.

```
is_active = equipment["status"] == "active"
has_inspection = equipment["inspection_complete"]
is_under_limit = (
  equipment["operating_hours"]
  <= equipment["maximum_hours"]
)

equipment["is_ready"] = (
  is_active and has_inspection and is_under_limit
)
```

Readable debugging pattern

- name the checks
- print the checks
- combine the checks
- store the result

Clear code is easier to verify.

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Guided exercise 2: project record

Calculate and add a derived value.

```
project = {
  "project_id": "PRJ-2401",
  "name": "Network Modernization",
  "completed_tasks": 18,
  "total_tasks": 24,
  "approved": True,
}

project["completion_percentage"] = (
  project["completed_tasks"]
  / project["total_tasks"]
  * 100
)
```

Then print:

- project name
- completion percentage
- approval status
- the whole dictionary

Derived values become part of the record.

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Guided exercise 3: missing keys

Predict before running.

```
print(project.get("budget"))
print(project.get("budget", 0))
print(project["budget"])
```

Prediction

1. None
2. 0
3. KeyError

Discussion

When should missing data be allowed?
When should it stop the program?

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Applied challenge: build the record

Convert standalone variables into one dictionary.

```
record_id = "INGEST-2401"
file_name = "equipment.csv"
file_size_mb = 18.5
record_count = 12450
error_count = 2
approved = True
```

```
record = {
  "record_id": record_id,
  "file_name": file_name,
  "file_size_mb": file_size_mb,
  "record_count": record_count,
  "error_count": error_count,
  "approved": approved,
}
```

Step one: gather related facts
under meaningful keys.

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Applied challenge: add derived fields

Now make the record useful.

```
record["supported_file_type"] = (
  record["file_name"].lower().endswith(".csv")
)

record["has_errors"] = record["error_count"] > 0

record["ready_for_processing"] = (
  record["supported_file_type"]
  and record["file_size_mb"] > 0
  and record["record_count"] > 0
  and record["approved"]
)
```

Derived fields answer questions:

- Is the file supported?
- Does it have errors?
- Can processing continue?
- What should a human see?

Readable keys make later code easier.

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Common dictionary mistakes

Most errors are small spelling or syntax problems.

Missing comma

```
record = {
  "id": "EQ-1"
  "status": "active"
}
```

Wrong key name

```
record["status"]
record["Status"]
```

Expecting missing key

```
record["budget"]
```

Debugging move: print(record) and print(record.keys()).

BLOCK 19

Mini knowledge check

Answer in plain language first.

1. What is the difference between a key and a value?
2. When should you use .get()?
3. Why is a dictionary easier to read than a list for records?
4. What does record["metrics"]["temperature"] mean?

Predict → run → explain.

BLOCK 19

Wrap-up: choosing a collection

Pick the structure that matches the data.

List	Tuple	Dictionary	Coming next
Many values identified by order	Fixed group with meaning by position	Labeled fields inside a record	Loops will automate work across collections.

```
record["ready"] = True
print(record["ready"])
```

Dictionaries make code read like the data it describes.